

Procedural Wrapping of Cloth-Like Strips on 3D Meshes

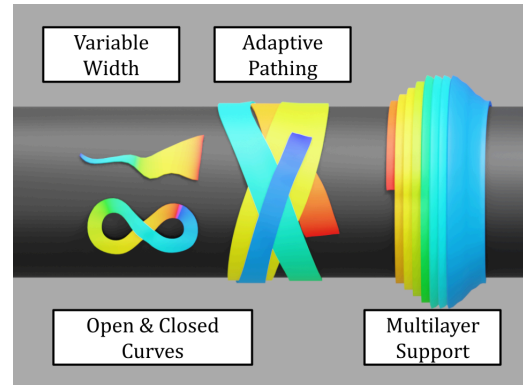
Sole Developer / Mar 2023 – Apr 2023

Overview:

This project developed a Geometry Nodes tool in Blender that converts curve objects into mesh strips and procedurally wraps them onto 3D surfaces with customizable layering, thickness, and variable width. Designed for intuitive, real-time adjustments using Blender's curve system, it supports both open and closed curves and prioritizes clean UV layouts for material application. The tool enables flexible, detailed shaping suited for applications such as bandage wrapping or other surface applications.

Tools & Skills:

Blender (Geometry Nodes), Procedural Modeling, 3D Geometry & Vector Mathematics, Node-Based Workflow Optimization, Visualization and Debugging Techniques



Demo of procedural wrapping with overlap, variable width, and curve flexibility.

Objective:

Create a Geometry Nodes system in Blender that transforms curve objects into mesh strips and wraps them onto 3D meshes with support for layering and thickness control.

Design Highlights:

- The tool allows for variable strip width along the curve, enabling flexible and detailed shaping.
- It supports multiple layers and controlled self-overlap, preserving consistent stacking behavior.
- Built on Blender's curve system, it enables intuitive, real-time adjustment of wrap paths
- Accepts open and closed (cyclic) curves to support continuous wrapping and diverse use cases..
- Prioritizes clean UV layout for materials and textures, with secondary support for cloth simulation and additional modeling..

Lessons Learned:

Maintaining a clean and well-organized node tree greatly improves the ease of making modifications, even after stepping away from the project for an extended period. Procedurally generating helper curves can introduce unexpected behavior during intermediate steps, highlighting the need for careful handling and debugging. Additionally, increasing the number of control points provides finer control over the resulting mesh but can clutter the editor, making selection and manipulation more difficult.

Next Steps:

Enhance the tool's ability to handle extreme curvature and closely positioned faces on complex target meshes. Implement twist and thickness controls along the strip's path to enable more precise shaping. Introduce visual markers and overlays to clearly indicate overlap order and width variation. Package the setup as a reusable asset to facilitate easier deployment across projects. Additionally, incorporate tooltips and in-graph annotations to improve usability and support onboarding for new users.